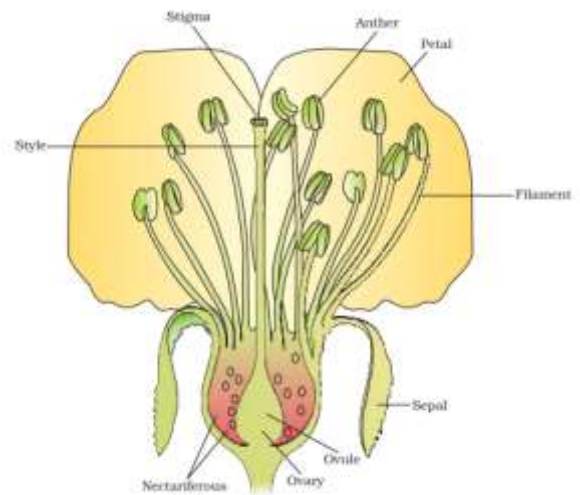


CHAPTER – 1**SEXUAL REPRODUCTION IN FLOWERING PLANTS****Characteristics of Angiosperms**

- They are commonly called **flowering plants** or **phanerogams**.
- They are commonly known as closed seed plants, because the ovules are covered by ovary.
- They are the most advanced group of plants both physiologically and anatomically.
- They are most advanced tracheophytes (both xylem and phloem are well developed).
- The members exhibit double fertilization.
- The endosperms are always triploid in nature.

Flower

- Flower is a highly condensed reproductive shoot of a plant or it is a special reproductive branch meant for sexual reproduction.
- The study of flower is called **Anthology**.

**Parts of a flower**

- Pedicel – It is the stalk of the flower.
- Thalamus/Receptacle – It is the tip of the pedicel becomes flattened or disc like holds or supports all the floral parts. Thalamus bears four floral parts. They are
 - a) Calyx**
 - It is the outermost floral part and green in colour.
 - It helps in photosynthesis and gives protection when flowers are in bud condition.
 - The individual units of calyx are called **sepals**.
 - b) Corolla**
 - It is the second floral whorls brightly coloured due to the presence of a pigment called **Anthocyanin**.
 - It helps in attraction of birds and insects for pollination.
 - The individual units of corolla are called **petals**.
 - c) Androecium**
 - It is the third floral whorl and called male reproductive part.
 - The individual units are called **stamens**.
 - Each stamen consists of anther, connective and filament.
 - d) Gynoecium**
 - It is the inner most floral whorl and called female reproductive part.
 - The individual units are called **pistil** or **carpel**.
 - Each pistil consists of ovary, style and stigma.

Note

- Both calyx and corolla are called Accessory whorls because they do not take part in sexual reproduction.
- Both Androecium and gynoecium are called essential or necessary whorls because they directly and actively take part in sexual reproduction.

Note: Perianth: a flower in which calyx and corolla are undifferentiated and is called perianth. Such a flower is called homochlamydeous flower. A unit or segment of perianth is called tepal. Eg: flower of champaka coconut, onion etc

Functions of flower

- It helps in production of pollen and ovules.
- It aids in pollination.
- It provides nectar to attract certain pollinator
- It helps in fertilization. After fertilization ovary is the one which develops into a fruit
- It helps the ovules to develop into seeds

Steps involved in Sexual Reproduction in flowering plants

1. Pre – fertilization events
2. Pollination
3. Double fertilization
4. Post – fertilization events

1. Pre – fertilization events

- a. Development and formation of male gametophyte (pollen grain)
- b. Development and formation of female gametophyte (embryo)

**a. Development and formation of male gametophyte (Pollen grain)
(Stamen, Microsporangium and Pollen grain)**

- A stamen is a male reproductive part or unit of a flower or Microsporophyll of a flower.
- It is made up of 2 parts. Namely anther and filament.
- The filament is a long slender stalk. Its proximal end is attached to the thalamus or petal of the flower.
- The anther is generally bilobed structure with each lobe having two theca i.e., they are **ditheous**.
- It has four-sided structure consisting of four microsporangia.
- The microsporangia develop further and become pollen sacs and are packed with pollen grains.

NOTE: A ditheous anther is **tetrasporangiate**, while monotheous stamen is **bisporangiate**.

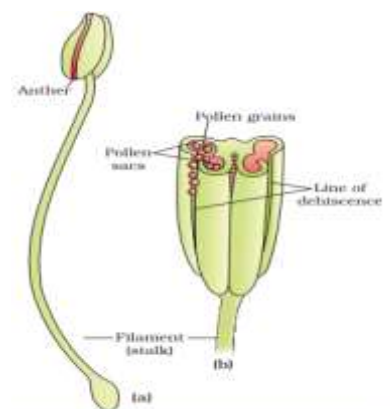
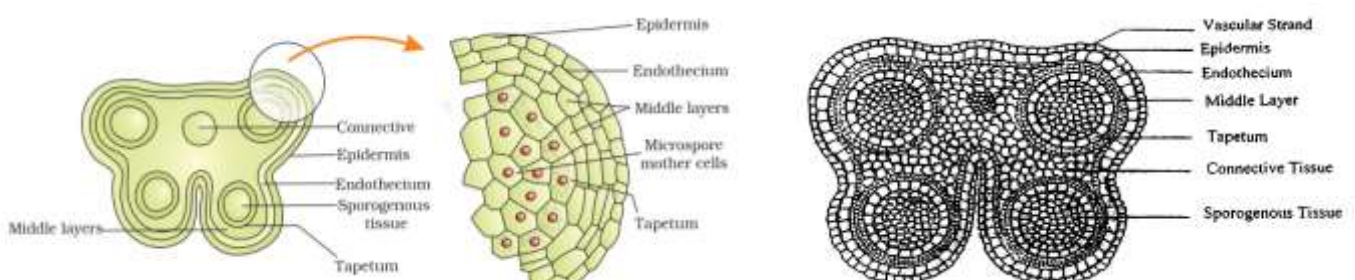


Figure 2.2 (a) A typical stamen; (b) three-dimensional cut section of an anther

Structure of Microsporangium

- The microsporangium consists of following parts.

a. Anther wall layers b. Connective c. Pollen grain

b. Anther wall layers

- They are derived from primary parietal cell by repeated anticlinal and periclinal division.
- There are four anther wall layers. They are,

1. Epidermis

- It is the outermost anther wall layer.
- It is a single layered made up of compactly arranged rectangular shaped thick-walled cells and gives protection.

2. Endothecium

- It is the second anther wall layer. It is also a single layered made up of compactly arranged cells.
- At the time of maturity, the endothelial cells produce fibrous thickenings made up of **cellulose** and **callose**.
- These fibrous thickenings are hygroscopic in nature i.e., they absorb moisture content and helps in dehiscence of anther.
- Some of the endothelial cells are modified into delicate cells called **stomium** where anther bursts.

3. Middle layer

- It is the third anther wall layer.
- It is made up of 2 to 3 layers of rectangular shaped compactly arranged parenchymatous cells.
- This layer stores reserve food materials and are **short lived (ephemeral)**.

4. Tapetum

- It is the innermost layer of anther wall layer made up of a single layered elongated cell.
- This layer is also called nutritive layer because it provides nourishment to the developing pollen grains.
- The cells of the tapetum contain dense cytoplasm and generally have more than one nucleus.
- A group of compactly arranged sporogenous cells are present in the young anther is called **sporogenous tissue**.

b. Connective

- At the junction between the two anther lobes specialized conducting tissues are present and are called connective.
- They provide nourishment to middle layer.

c. Pollen grain

- It is the male gametophyte derived from primary sporogenous cell by meiotic division.

Functions of Tapetum

- It is a chief nutritive layer for developing pollen grains.
- It induces meiotic stimulus to sporogenous cells.
- It produces a highly resistant biological chemical called **Sporopollenin**, which is responsible for formation of exine.
- It also produces pollen kit (It is a lipid molecule protects the pollen grain from dehydration).
- It also produces specialized nutritive bodies called **Ubisch bodies**.

Additional Points for Competitive Examination

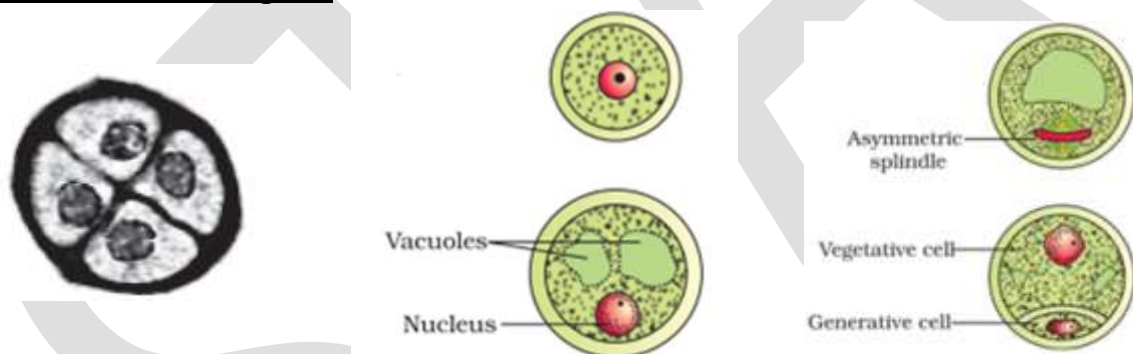
Tapetum is of two types:

- i) **Secretory or glandular tapetum:** These cells secrete sporopollenin, pollenkit and compatibility proteins. These cells provide Ubisch bodies which help in the ornamentation of exine as they have a chemical called sporopollenin which is deposited on them.
- ii) **Amoeboid tapetum:** Cells undergo breakdown and their entire protoplast move towards centre to nourish **microspores**.

Microsporogenesis

- The process of formation of microspores from a **pollen mother cell (PMC)** through meiosis or the process of formation of microspores or pollen grains from diploid primary sporogenous cell is called **microsporogenesis**.
- The primary sporogenous cell undergoes repeated meiotic division to form number of sporogenous cell.
- Each sporogenous cell gets meiotic stimulus to form **microspore mother cell (MMC)**.
- Each diploid microspore mother cell undergoes meiotic divisions to form four microspores or pollen grains.

Structure of a mature Pollen grain



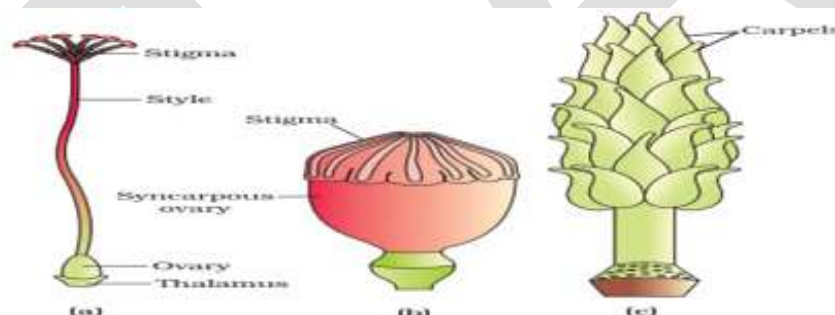
- The male gametophyte in angiosperm is called **pollen grain**.
- It is haploid in nature, spherical in shape and measuring **25 to 50 microns in diameter**.
- It is covered by two layers. Namely **Exine** and **Intine**.
- Exine** is the outer layer made up of highly resistant chemical called **Sporopollenin**.
- It can withstand high temperature, strong acids and alkalis. There is no known enzyme which can digest sporopollenin.
- Exine layer is discontinuous which leads to the formation of small aperture called **pollen pore** or **germ pore** through which pollen tube germinate.
- The inner layer is called intine which is thin and made up of cellulose and pectin (pectocellulose).
- The cytoplasm of pollen grain is surrounded by a plasma membrane.
- When the pollen grain is mature it contains two unequal sized cells, namely the bigger vegetative cell (tube cell) and smaller generative cell.
- Vegetative cell remains in the center and has reserve food materials and large irregularly shaped nucleus.
- The generative cell is spindle shaped with dense cytoplasm and a nucleus present at the periphery.
- In over **60% of angiosperms**, pollen grains shed at **2 cell stage**.
- In over **40% of angiosperms**, the generative cell divides by mitosis to form 2 male gametes and shed at **3 cell stage**.
- During germination, the intine grows out as pollen tube through one of the germ pores in the exine.

NOTE

- Pollen grains of many species cause severe allergies and bronchial afflictions in some people often leading to chronic respiratory disorders like asthma, bronchitis etc.
- It may be mentioned that **Parthenium** or **carrot grass** that came into India as a contaminant with imported wheat, has become ubiquitous in occurrence and causes **pollen allergy**.
- Even the large number pollen grains can be preserved for many years in liquid nitrogen at -196°C . Such stored pollen grains can be used as **pollen banks** in crop breeding programmes.
- In some cereals such as **Rice** and **Wheat** pollen grains lose viability within **30 minutes** of their release.
- In some members of **Solanaceae**, **Rosaceae** and **Leguminosae** the viability of pollen grains maintains for **one month**.

Economic Importance of Pollen grain

- Pollen grains are rich in nutrients.
- In recent years it has become a fashion to use pollen tablets as food supplement.
- In Western countries, a large number of pollen products in the form of tablets and syrups are available in the market.
- Pollen consumption is believed to increase performance of athletes and race horses.

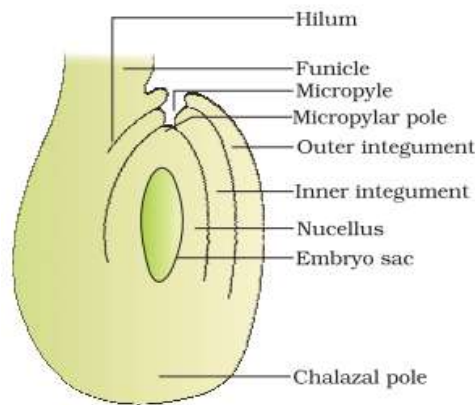
c. Development and formation of female gametophyte (Embryo sac)**THE PISTIL, MEGASPORANGIUM (OVULE) AND EMBRYO SAC**

- The gynoecium (Megasporophyll) represents the female reproductive part of the flower.
- The gynoecium may consist of a single pistil (monocarpellary) or may have more than one pistil (multicarpellary).
- When there is more than one, the pistils may be fused together (syncarpous).
- Each pistil has three parts, the stigma, style and ovary.
- The stigma is the terminal part of the style which is usually sticky and receives pollen grains during pollination.
- The style is the elongated slender part beneath the stigma.
- The ovary bulged part of the pistil is the ovary. Inside the ovary is the ovarian cavity (locule).
- The placenta is located inside the ovarian cavity. From the placenta megasporangia arises which is commonly called **ovules**.
- The number of ovules in an ovary may be one (wheat, paddy, and mango) to many (papaya, water melon, orchids).

Types of Ovary

- Monocarpellary ovary:** A gynoecium with a single carpel or pistil. **E.g.:** - Pea.
- Multicarpellary:** A gynoecium having more than one carpel or pistil. It is of following types.
 - Bicarpellary ovary:** A gynoecium with two carpels. **E.g.:** - Vinca rosea.
 - Tricarpellary:** - A gynoecium with three carpels. **E.g.:** - Castor.
- Apocarpous ovary:** A gynoecium consisting of free carpels i.e., not fused. **E.g.:** - Michelia champaka.
- Syncarpous ovary:** A gynoecium consisting of fused carpels **E.g.:** - Lady's finger, Papaver.

Structure of L.S of mature Anatropous Ovule/ The Megasporangium (Ovule)



A diagrammatic view of a typical anatropous ovule

- It consists of three parts. Namely, **Funicle**, **Ovule body** and **Embryo sac**.

a. Funicle

- The stalk of the ovule is called **funicle**.
- The funicle winds the ovule body to the placenta.
- The point of attachment of funicle to the ovule body is called **Hilum**.
- Hilum consisting of a nutritive strand called **Raphae**.

b. Ovule body

- The base of the ovule body is called **Chalaza**.
- The entire ovule body is filled with nutritive tissues which is made up of compactly arranged paranchymatous cells called **Nucellus**.
- Nucellus is covered by two protective coverings called **inner** and **outer integuments**.
- At the free end of the ovule body the two integuments leave a minute microscopic canopy is called **Micropyle**.

c. Embryo sac

- It is the female gametophyte located in the nucellus.
- An ovule generally has a single embryo sac formed from a megaspore through reduction division.
- A mature embryo sac has **seven cells** and **eight nuclei**.
- Out of these seven cells, three haploid antipodal cells are present towards chalaza and three haploid cells are called Egg apparatus and are present towards Micropyle (**egg apparatus consists of one haploid functional egg cell and two haploid non-functional synergid cells**) and one central cell having diploid secondary nucleus.
- The degenerated part of the nucellus in the ovule is called **Perisperm** or the residual, persistent nucellus is called **Perisperm**. **E.g.:** - black pepper, beet, coffee etc.

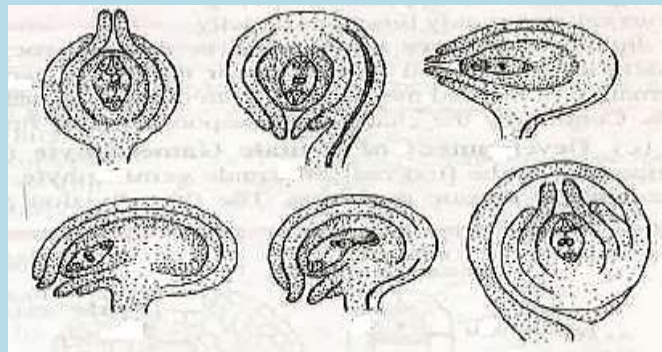
Additional Points for Competitive Examination

Tapetum is of two types:

Types of ovule on the basis of integuments

- A single integumented ovule is called unitegmic ovule – example – members of Gamopetalae and Gymnosperm.
- Two integumented ovule is called bitegmic ovule. Example – In most of Angiosperm [Polypetalae – Capsella and Monocots].
- The ovule in which integuments are absent is called Ategmic ovule e.g. Olax, Liriosma, Loranthus and Santalum.

- **Depending upon the development of nucellus, ovules are of two types:**
 - i. Crassinucellate ovule: The nucellus is well developed, e.g. polypetalae
 - ii. Tenuinucellate ovule: The nucellus is poorly developed, e.g. gamopetalae
- **Types of ovules** – On the basis of the relative position of micropyle, body of the ovule and funicle, i.e., the degree of curvature, there are six types of ovules:
- **Orthotropous:** The micropyle, chalaza and funicle are in a straight line. This is the most primitive type of ovule e.g. piper, polygonum, cycas
- **Anatropous:** The ovule turns at 180° angle. Thus, it is inverted ovule. Micropyle lies close to hilum or at side of hilum, e.g., it is most common ovule and found in 82% of angiosperm families.
- **Hemi anatropous:** Ovule turns at 90° angle upon the funicle or body of ovule is at right angle to the funicle, e.g. Ranunculus.
- **Campylotropous:** Ovule is curved more or less at right angle to funicle. Micropylar end is bend down slightly, e.g., in members of Leguminosae, Cruciferae
- **Amphitropous:** Ovule as well as embryo sac is curved like horse shoe, e.g., Lemna, Poppy, Alisma
- **Circinotropous:** The ovule turns at more than 360° angle, so funicle becomes coiled around the ovule, e.g., Opuntia (Cactaceae) Plumbaginaceae.



Additional Points for Competitive Examination

Special Integuments:

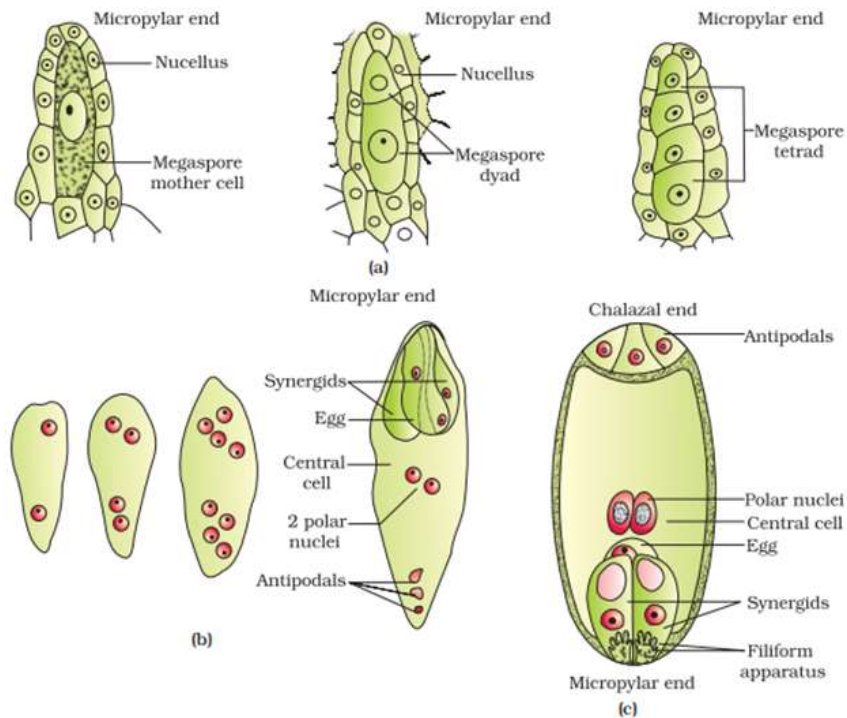
- Aril – It is the type of third integument which develops from funicle at the base of the ovule e.g. Myristica, Asphodelus and Litchi
- Arillode – It develops from the tips of the outer integument and grows downwards and surrounds the entire ovule e.g. Pithecolobium (Inga dulce)
- Sarcotesta – When outer integument or testa becomes fleshy then it is called sarcotesta e.g. Magnoliaceae (Magnolia)
- Caruncle or Strophiole – It is formed due to the proliferation (out growth) of outer integuments over the micropyle e.g. Ricinus cumunis (castor). It is made up of sugary contents, so it helps in absorption of water during germination of seeds and dispersal of seeds occurs by ants which is called myrmecochory
- Coma – In some of the plants unicellular filaments like structures are present on the seed which is formed by cells of outer surface of outer integument and present on testa of seeds. Such seeds are known as “Comose seeds”. E.g. Calotropis and Gossypium (Cotton).

Megasporogenesis

- The process of formation of megaspores from megaspore mother cell (MMC) is called **megasporogenesis**.
- Ovules generally differentiate a single megaspore mother cell (MMC) in the micropylar region of the nucellus.

- It is a large cell containing dense cytoplasm and a prominent nucleus.
- The MMC undergoes meiotic division which results in the production of four megaspores.

Female gametophyte/Embryo sac



- In a majority of flowering plants, one of the megaspores is functional while the other three degenerate.
- Only one functional megaspore develops into the female gametophyte (embryo sac).
- The method of embryo sac formation from a single megaspore is termed as **monosporic development**.
- The haploid nucleus of functional megaspore divides mitotically thrice to form 8 nuclei.
- Out of these 8 nuclei, 4 nuclei migrate towards one pole of the embryo sac and the remaining 4 nuclei migrate towards opposite pole of the embryo sac.
- One nucleus from each pole migrates to the centre of the embryo sac and is called **polar nuclei**. These fuse to form secondary nucleus i.e. a **diploid secondary nucleus**.
- The 3 nuclei at the base of the embryo sac form the **antipodal cells** and the other 3 haploid nuclei at the apex of embryo sac form the **egg apparatus** consisting of a **central egg cell** and 2 **lateral synergids**.
- Thus, a typical angiosperm embryo sac at maturity has seven cells with eight nuclei.
- The synergid cell wall forms a highly thickened structure called the **Filiform apparatus**, which plays an important role in guiding the pollen tube to penetrate into the synergid.

Differences between Microsporogenesis and Megasporogenesis

Microsporogenesis	Megasporogenesis
It is the process of formation of microspores from a pollen mother cell	It is the process of formation of megaspore from a megaspore mother cell
It takes place inside the anther (pollen sac)	It takes place inside the ovules
Microspore mother cell gives rise to four haploid pollen grains	Megaspore mother cell gives rise to one embryo sac
Microsporogenesis is controlled by tapetum	Megasporogenesis is controlled by nucellus
All the four microspores of a spore tetrad are functional.	Only one megaspore of a spore-tetrad is functional.
Large numbers of microspore mother cells are functional in a microsporangium.	Generally, a single microspore mother cell is functional in a megasporangium

Pollination

- The transfer of pollen grains from the anther on to the stigma is called **pollination**.
- Pollination in angiosperms generally takes place at 2 celled stage i.e. pollen having vegetative cell (tube cell) and generative cell.
- If pollination occurs in closed flowers or the flowers which do not open at all is called **cleistogamy** and the flowers are called **cleistogamous flowers**.
- If pollination occurs in opened flowers i.e., the anthers and stigma are exposed, it is called **chasmogamy** and the flowers are called **chasmogamous flowers**. E.g.: -Viola (common pansy), Oxalis, Commelina

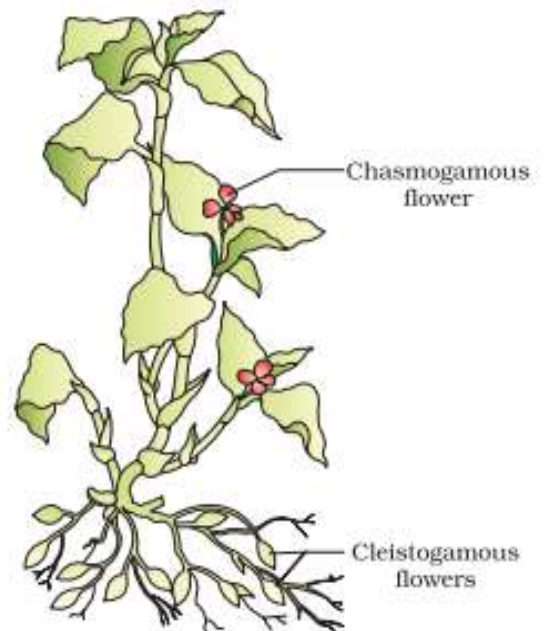
Advantage of Cleistogamy :

- It ensures seed formation even in the absence of any pollinating agent.
- It ensures pure line breeding of the plants which is essential for the studying genetics and heredity
- The plant doesn't need to produce scent, nectar or extra pollen to attract pollinators.

Disadvantage of Cleistogamy: The offsprings produced have limited genetic diversity.

Chasmogamous flowers are with exposed anthers and stigma, hence autogamy to occur requires synchrony in pollen release and stigma receptivity.

These are large scented colourful flowers and have nectar content in order to attract the pollinators.

**Kinds of Pollination**

Pollination is of two types. Namely

1. Self-pollination/autogamy
2. Cross pollination/Allogamy.

1. Self-pollination/Autogamy

- The transfer of pollen grains from anther to stigma of the same flower is called **self-pollination**.
- It takes place in only in those plants bearing bisexual flowers.
- It is of two types. Namely,
 - Homogamy:** - In this case both anthers and stigma of a bisexual flower mature at the same time. E.g.: - Wheat, Rice, Potato.
 - Cleistogamy:** - In this case the bisexual flowers never open but always remain closed so that self-pollination can take place. E.g.: - Ground nut, Pea, Balsam, Commelina.

2. Cross pollination/Allogamy

- The transfer of pollen grains from anther to stigma of another flower is called **cross pollination**.
- It occurs in all unisexual flowers and in some bisexual flower.
- It is of two types. Namely,
 - Geitonogamy:** - In this case cross pollination takes place between two flowers of the same species born on the same plant. E.g.: - Maize,

- It is functionally cross pollination and requires pollinating agent but genetically they are similar to autogamy since the pollen grains come from the same plant.

b. Xenogamy: - In this case cross pollination takes place between two flowers of the same species born on different plants.

Significance of xenogamy

- It brings genetically different types of pollen grains to the stigma, thus induces genetic variation.
- **Example:** Papaya plants exhibit xenogamy because, unisexual flowers borne on two different plants.

Agents of Cross Pollination

- Since the pollen grains are non-motile the transfer of pollen grains from the anther of one flower to the stigma of another flower requires the assistance of external agents.
- These agents are classified into two main categories. Abiotic and Biotic agents.
- The abiotic agents include non-living agents like wind, water.
- The biotic agents include living organisms such as bats, birds, snails, slugs.

Abiotic agents of cross pollination

1. **Anemophily:** - The pollination that occurs through the agency of wind. Such flowers are called **anemophilous flowers**. **E.g.:** - Grasses, Maize, Sugarcane etc.

Factors influencing anemophilous pollination

- The flowers are small, inconspicuous, and colourless.
- They do not produce scent and any nectar.
- The stamens have long filaments so that the anthers are freely exposed to wind.
- The pollen grains are small in size, very light, smooth, dry and powdery, so that they can easily carried by wind.
- Pollen grains are produced in large quantities so that at least some of them reach the flowers.
E.g.: - a) Cannabis produces 5, 00,000 pollen grains.
b) Tassel of maize give rise to 20-25 million pollen.
- Stigma is large, often lobed and feathery, so that the pollen grains in air can be trapped easily.
- Flowers often have a single ovule in each ovary and numerous flowers packed into an inflorescence.

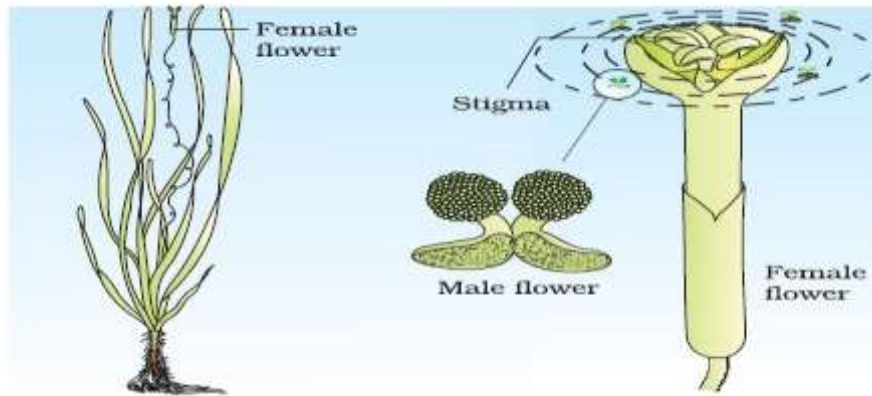
2. **Hydrophily:** - The pollination that occurs through the agency of water. Such flowers are called **hydrophilous flowers**. **E.g.:** -Hydrilla, Vallisneria, Zostera.

Factors influencing hydrophilous pollination

- Flowers are small, inconspicuous, and colourless.
- They do not produce scent and nectar.
- Quite rare in flowering plants
- Regular among the lower plant groups such as algae, bryophytes and pteridophytes.
- Calyx, corolla and other floral parts are unwettable.
- Pollen grains are small, non-sticky and coated by wax.
- Stigma is long unwettable and sticky.

Types of Hydrophily: - It is of two types. Namely, a. Epihydrophily b. Hypohydrophily

- a. **Epihydrophily:** - When pollination occurs on the surface of the water is called **Epihydrophily**. It occurs in Vallisnaria, Hydrilla.



- **Vallisnariaspinalis (Ribbon weed)** is a rooted submerged plant.
- It is unisexual or dioecious.
- The male flowers are small and at maturity they detach from the parent plant and float on the surface of water.
- The female flowers are large and at maturity they are brought to the surface of water by its long and slender stalk.
- The floating male flowers come close to the female flowers and the anthers burst open, the pollen grains get attached to the stigma effecting pollination.
- It is carried passively by water current; some of them eventually reach the female flower and the stigma.
- Immediately after pollination the long stalk of the female flower coils spirally and pulls it down into the water and development of fruit occurs only under water.

- b. **Hypohydrophily:** - When the pollination occurs inside the water is called **hypohydrophily**. It occurs in Ceratophyllum, Zostera

Pollination in Ceratophyllum/Hornwort/Coontail

- It is a fresh water submerged aquatic plant.
- It is bisexual plant.
- The same plant produces both male and female flowers.
- At maturity the anthers detach from the male flowers at the base. They move upward, reach the surface of water then break open to release pollen grains.
- The pollen grains are round without exine.
- The liberated pollen grains sink in water and come in contact with long and sticky stigmas of female flowers to achieve pollination.

Pollination in Zostera

- It is a marine submerged plant.
- It produces elongated needle like pollen grains without an exine.
- When pollen grains reach the stigma, they coil around it and bring about pollination.

Biotic agents of Cross Pollination

1. **Zoophily:** - The pollination that occurs by animals is called **zoophily**. It is of following types.

- a. **Entomophily:** - The pollination that occurs by insects. **E.g:-** Butterflies, Bees, Moths

Factors influencing Entomophilous pollination

- The flowers are brightly coloured.
 - The flowers emit scent to attract insects.
 - The flowers produce sugary juice called nectar, by **nectiferous glands** to attract insects for feeding.
 - The pollen grains are large, spiny and sticky so that they can get easily attached to the body of visiting insects.
 - The stigma becomes sticky to catch the pollen grains.
 - **Pollen/Nectar Robbers:** The floral visiting insects which may consume pollen or the nectar without bringing out pollination are referred to as pollen/nectar robbers.
- In some species floral rewards are provided for pollinators. Nectar, pollen grains, shelter, edible floral parts and young seed are the floral rewards.

E.g.: - a. In *Amorphophallus* – the tallest flower provides safe place to lay eggs.

- In *Yucca*, Pronuba moth bores a hole in the ovary and lay its eggs in it. Then it collects pollen from several flowers and pushes the whole of pollen down the hollow end of the stigma. Fertilization takes place and seeds develop. Larvae feed on developing seeds. Some seeds remain unconsumed to propagate the species.

b. Ornithophily: - The pollination that occurs by birds. **E.g.:** - Humming bird, Sun birds, Honey eaters

Factors influencing Ornithophilous Pollination

- The flowers are large and brightly coloured.
- They produce scent.
- They produce large quantities of nectar.

c. Chirapterophily: - Pollination by means bats. **E.g.:** - *Caesalpinia*

d. Malacophily: - Pollination by means snails and slugs. **E.g.:** - Grasses

e. Myrmecophily: - Pollination by means ants. **E.g.:** - Mango, Acacia

f. Cantharophily: - Pollination by means of beetles. **E.g.:** - Magnolia

Advantages of self-pollination

- Chances of pollination are more.
- Maintains purity of race and avoids mixing.
- Need not to be producing a large number of pollen grains.
- Flowers need not possess devices such as large showy petals, presence of scent and nectar to attract insects.

Disadvantages of self-pollination

- Progeny continuously gets weakened after every generation.
- Less chances of production of new species and varieties.

Advantages of Cross-pollination

- Sometimes harmful characters are eliminated.
- It reaches the production of new varieties.
- It increases variability; certain new and useful characters appear in the off springs.
- It also increases the power of adaptation among the off springs.
- The number of seed production can be increased among useful cereal crops.
- Several disease resistant varieties of important crop plants have been produced.

Disadvantages of Cross-Pollination

- It is uncertain because a factor of chance is always involved.
- Because of the distance barrier pollination may fail.
- It is not economical. Because the plants waste a lot of food materials and energy in unnecessary adaptation and devices.
- It involves addition of some undesirable characters and loss of some important characters

Differences between Self-pollination and Cross-pollination

Self-Pollination	Cross Pollination
It always occurs through touch or wind	It always occurs through an external agent
Transfer of pollen grains from anther to stigma of same flower	Transfer of pollen grains from anther to stigma of different flowers
It always takes place in bisexual flowers	It takes place in both unisexual and bisexual flowers
Both anthers and stigmas mature simultaneously	The anthers and stigmas mature at different time
It is due to homogamy	It is due to heterogamy or dichogamy
It gives rise to pure line	It gives rise to off springs having variation and hybrids
It preserves parental characters	It does not preserve parental characters
Origin of new species is not possible	It produces new races and varieties

Out breeding devices/Contrivances for cross pollination

- Majority of angiosperms produce bisexual flowers and such flowers are likely to get self-pollinated.
- Continuous self-pollination results in inbreeding depression (suitable characters decline).
- To avoid self-pollination and to encourage cross pollination the flowering plants have developed many devices. They are called **out breeding devices**.
- Some of the out-breeding devices are as follows.

1. **Unisexuality/decliny:** It is a condition where plants produce only unisexual flowers. This prevents self-pollination and encourage cross pollination.

E.g.: -

a. Castor and Maize are **monoecious** where unisexual male and female flowers borne on same plant prevents autogamy but not geitonogamy.

b. In several species like **papaya**, male and female flowers present on different plants (**dioecious**). This condition prevents both autogamy and geitonogamy.

2. **Dichogamy/Heterogamy:** It is a situation where in a bisexual flower the male and female sex organs mature at different time to avoid self-pollination. There are two types of heterogamy. They are

a. Protandry: It is a condition, where in a bisexual flower the anthers mature earlier before the stigma. Such flowers are called **protandrous**. **E.g.:** - Helianthus, Leucas aspera, Cotton, Sunflower

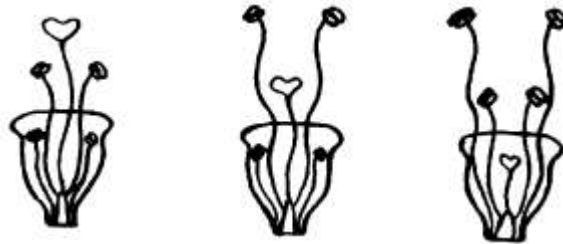
b. Protogyny: It is a condition, where in a bisexual flower stigma mature earlier before anthers. Such flowers are called **protogynous**. **E.g.:** - Mirabilis, Ficus,

3. **Self-sterility/ Incompatibility:** It is a condition where even if self-pollination occurs; the pollen grains fail to germinate on the stigma of same flower due to mutual inhibition. In such plants seeds are

produced only after cross pollination. Or the phenomenon in which the pollen grains fail to fertilize the egg of same flower is called **self-sterility** or **incompatibility**.

E.g.: -Brssica

- 4. Heterostyly:** It is a condition where in some bisexual flowers the stamens and styles are of different length. This situation avoids self-pollination and facilitates cross pollination. **E.g.:** - Jasmine, Oxalis



- 5. Herkogamy:** It is condition where a physical barrier appears between male and female sex organs. Barrier may be in the form of a hood of covering the anther or stigma. **E.g.:** -Vincarosea, Iris, Calotropis.

E.g.: - In *Calotropis*, pollens are present in a sac like structure called pollinium and pollination is at the mercy of Insects

Pollen Pistil interaction

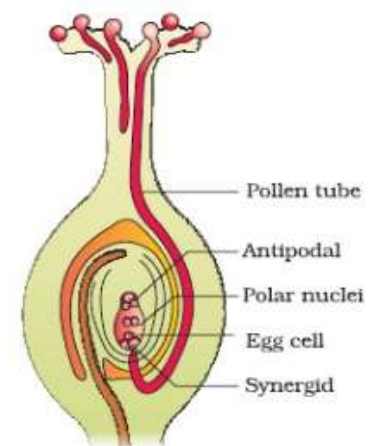
- It is one of the important events takes place in flowering plants before fertilization.
- The events from deposition of pollen grains to the stigma until the entry of pollen tube into the ovules are referred to as **pollen-pistil interaction** or after pollination chemical reaction or interaction between pollen grains and pistil of gynoecium within the flower or between the flowers due to some chemicals present in them is called **pollen-pistil interaction**.
- Pollination does not guarantee the transfer of the compatible pollen of the same species as the stigma. Often, pollen of the wrong type, either from other species or from the same plant even if it is self-incompatible also deposit on the stigma.
- The pistil has the ability to recognize the pollen of its **compatible** or **incompatible type**.
- If it is compatible type, the pistil accepts the pollen and promotes post-pollination events that lead to fertilization.
- If the pollen is incompatible, the pistil rejects the pollen by preventing pollen germination on the stigma or the growth of pollen tube in the style.
- Acceptance or rejection between pollen grain and the pistil is mediate by the chemical components.
- The pollen grain germinated on the stigma to produce a pollen tube one of the germ pores.
- The contents of the pollen grain move into the pollen tube. Pollen tube grows through the tissues of the stigma and style and reached the ovary.
- In few plant species, the pollen grains are shed at two celled stage (a vegetative cell and a generative cell). In such plants, the generative cell divides and forms the two male gametes during the growth of the pollen tube.
- In other plant species where pollen grains are shed at three celled stages, pollen tube carries the two male gametes.
- Pollen tube, after reaching the ovary, enters the ovule through micropyle and then enters one of the synergids through the filiform apparatus. Recent studies have shown that filiform apparatus present at the micropylar region guides the entry of pollen tube.

Artificial Hybridization

- It is a cross fertilization done by plant breeders.
- It is one of the approaches of crop improvement programmes.
- It is a process of crossing of two plants with different characters to produce commercially a superior hybrid with desirable characters. It is one of the major approaches in crop improvement.
- In this technique only the desired pollen grains are used for pollination and the stigma is protected to prevent contamination from unwanted pollen.
- It can be achieved by two processes. Namely, **Emasculation** and **Bagging techniques**.
- **Emasculation** is a process of removing of anthers in bud condition from the bisexual flowers by using a pair of forceps without affecting female reproductive organs or by immersing the inflorescence in hot water with 40-45 °C for about ten minutes. This plant is used as female parent for artificial hybridization.
- The emasculated flowers are covered with a bag made up of butter paper or polythene bag having minute pores for aeration. The process is called **bagging**.
- When the stigma of emasculated flowers mature, bag is removed and the desired pollen grains from male plants are dusted on the stigma using brush. The pollinated flowers are rebagged till fruit develops.
- Emasculation is not required for unisexual flowers. However, flowers are bagged to prevent contamination of its stigma with unwanted pollen or undesired pollination.

Double Fertilization

- The process of fusion of male gamete with the female gamete is called fertilization or syngamy.
- After pollination, the pollen grains on the stigma begin to germinate.
- The pollen grains are shed at two celled stage (a vegetative cell and a generative cell).
- The exine of the pollen grain ruptures and the intine comes out through the germ pore in the form of a long narrow pollen tube. The generative cell divides and gives rise to two male gametes.
- The pollen tube elongates and reaches the ovule through **micropyle (Porogamy)**. It is the most common method or through **chalazal end (Chalazogamy)** or through **integuments (Mesogamy)**.
- Inside the embryo sac the tip of the pollen tube breaks and the two male gametes are released.
- One male gamete fuse with the egg cell to form **diploid zygote**. This is known as **first fertilization**.
- Second male gamete fuses with two polar nuclei or secondary nucleus to form **Triploid Primary Endosperm Nucleus (PEN)**. As this involves the fusion of three haploid nuclei and it is called **Triple fusion**.
- Since the fertilization occurs twice and process is called **Double fertilization**.
- This phenomenon was first reported by **S.G Nawaschin (1898)** in **Lilium** and **Fritillaria** species and later confirmed by **Guignard** in **1899**.
- After fertilization, the **zygote** develops into **embryo** and **PEN** develops into Endosperm.



Longitudinal section of a flower showing growth of pollen tube

Significance of double fertilization

- Fertilization restores diploid number of chromosomes in the zygote.
- The diploid zygote develops into an embryo.

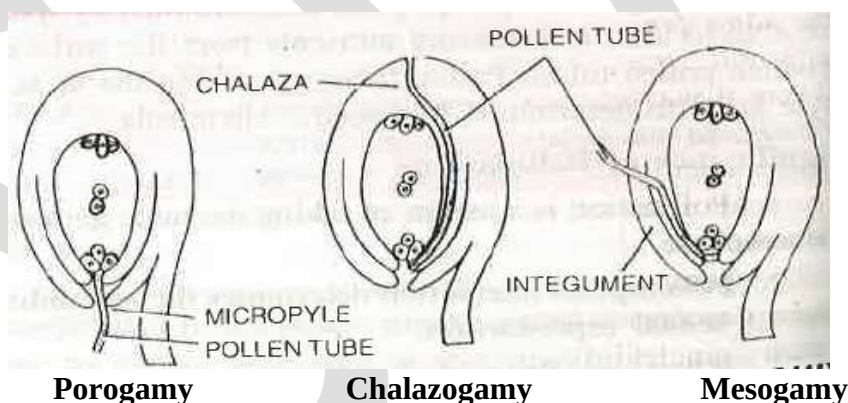
- The PEN develops into endosperm to provide nutrition to the developing embryo.
- The ovule develops into seed and the ovary develops into fruit after fertilization.
- It brings about genetic variations in those plants which exhibit cross pollination.

NOTE: Soon after the process of fertilization, the following changes will take place in a flower by which the flower is replaced by a fruit.

<i>Calyx</i>	–	generally falls off
<i>Corolla</i>	–	falls off
<i>Stamens</i>	–	falls off
<i>Style and Stigma</i>	–	shrivel and wither away
<i>Integuments of ovule</i>	–	become seed coats.
<i>Micropyle</i>	–	remain as pore in the testa
<i>Synergids and Antipodals</i>	–	degenerate
<i>Nucellus</i>	–	absorbed or left as perisperm
<i>Zygote</i>	–	develops into embryo
<i>Primary endosperm nucleus in central cell</i>	–	develops into endosperm
<i>Ovule</i>	–	develops into seed
<i>Ovary</i>	–	develops into fruits

NOTE

- The entry of pollen tube into the ovule through chalaza is called **Chalazogamy**.
E.g.: - Casuarina
- The entry of pollen tube into the ovule through integuments is called **Mesogamy**.
E.g.: - Cucurbita
- The entry of pollen tube into the ovule through micropyle is called **Porogamy**.



Post-Fertilization Events

- After fertilization certain changes are observed in the flower.
- These changes include endosperm formation, embryo formation, seed formation and fruit formation.
- All these events are collectively called post-fertilization events.

Endosperm

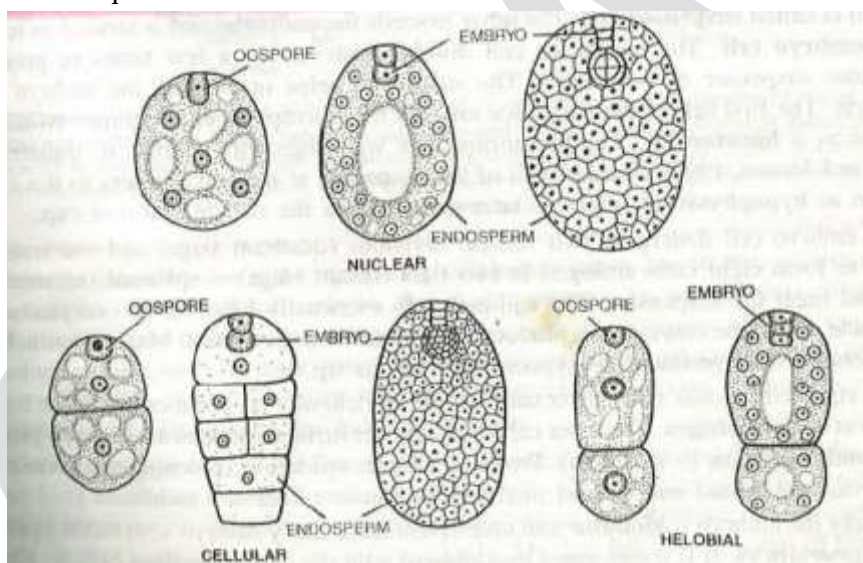
- It is a nutritive tissue which provides nourishment to the developing embryo.
- It is a Triploid structure formed by mitotic divisions of **primary endosperm nucleus (PEN)**.
- Depending upon the mode of its formation, Angiospermic endosperm is mainly of three types.
 - a. Nuclear endosperm
 - b. Cellular endosperm
 - c. Helobial endosperm

a. Nuclear endosperm

- It is the most common type of endosperm found in several angiosperms.

- In this type, PEN under goes repeated divisions without wall formation producing a large number of free nuclei.
- A vacuole appears in the Centre and pushes the cytoplasm containing the nuclei to the periphery.
- Later wall formation takes place around the nuclei to form multicellular endosperm. **E.g.:** - Maize, wheat, rice.
- In coconut there is an outer white kernel multicellular solid endosperm and the centre is filled with free nuclear endosperm called **coconut water** or **coconut milk**

1. **Nuclear type:** In this type of endosperm, the repeated mitotic division of PEN is not accompanied by wall formation. The nuclei produced are free in the cytoplasm of the embryo sac. They may remain free indefinitely or wall formation takes place later.
2. **Cellular type:** In this case, each division of PEN is followed by cytokinesis, hence forms a triploid endosperm tissue accompanied by wall formation. The cells of this tissue are filled with reserve food materials and are used for the nutrition of developing embryo.
3. **Helobial type:** It is an intermediate type. The first nuclear division is accompanied by cytokinesis but the subsequent ones are free nuclear.



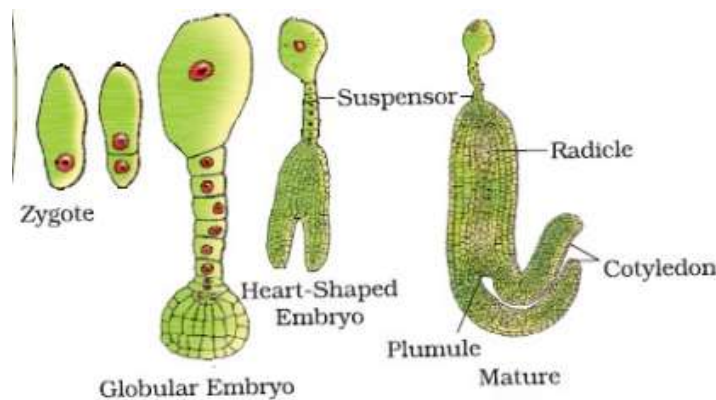
Fate of endosperm

- As the endosperm develops, it fills up the nucellus.
- In some seeds the developing embryo completely consumes endosperm. Such seeds without endosperm are known as **non-endospermic seeds** or **ex-aluminous seeds**. **E.g.:** - Pea, bean, groundnut.
- In some seeds the developing embryo do not consumes the endosperm completely. Such seeds persist even at maturity are known as **albuminous seeds** or **endospermic sees**. **E.g.:** - Castor, Coconut, wheat, maize, barley, sunflower

Embryo

- After fertilization, the fertilized egg or zygote develops into an embryo.
- The process of development of mature embryo from diploid zygote is called **embryogenesis**.
- In the early stages of development, the embryo of dicots and monocots are similar, but in later stages they differ considerably.
- The development of embryo is endospermic i.e. towards inside.

Structure of Dicot embryo/Stages involved in the development of Dicot embryo

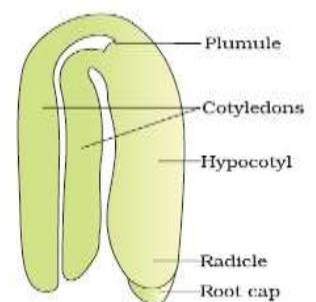


Stages in the development of a Dicot Embryo

- The zygote divides transversely and produces a two celled **proembryo** of unequal sized cells.
- The **larger cell** lying towards micropylar end is the **basal cell** or **suspensor cell** and the **smaller cell** lying towards chalazal end is the **terminal cell** or **embryo cell**.
- The suspensor cell undergoes a number of transverse divisions to form a filament of 6 to 10 **suspensor cells**.
- The first cell of the suspensor lying towards micropylar end becomes swollen to form **Haustorium**.
- Last cell of the suspensor functions as **hypophysis** and later gives rise to **radicle** and **root cap**.
- The embryo cell undergoes longitudinal divisions twice and at right angles to each other to form **4 cells**. These 4 cells then divide to form **8 cells** or **octet stage**.
- Out of these 8 cells the **lower 4 cells** give rise to **plumule** and 2 **cotyledons** while the **upper 4 cells** give rise to **radicle** and **hypocotyl**.
- All the 8 cells undergo periclinal divisions to form **outer single layered epidermal cells** called **dermatogen** and **inner layers of promeristem** and **procambium**.
- Initially the embryo is **globular** later with the growth of cotyledons it becomes heart shaped and finally the suspensor cells degenerate and mature embryo is formed.
- This type of dicot embryo development is called as **Crucifer type/Onagrad type**.

Structure of Dicot embryo

- A typical dicot embryo consists of an embryonal axis and two cotyledons.
- The portion of embryonal axis above the level of cotyledons is the **epicotyl**, which terminates with the **plumule (stem tip)**.
- The cylindrical portion below the level of cotyledons is the **hypocotyl** that terminates at its lower end in the **radicle (root tip)**.
- The root tip is covered with a **root cap**.

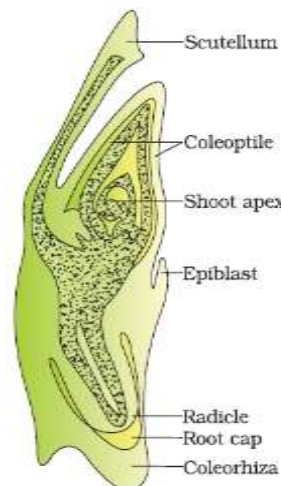


Structure of Dicot seed

- All dicotyledonous seed consists of the following parts.
 - a. Seed coat:** It consists of two layers namely, an **outer thick layer** called **testa** which is impermeable to water. The **inner thin layer** called **tegmen** which is permeable to water.
 - b. Embryo:** In dicot seed the embryo consists of the following parts.
 - i. Embryonal axis:**
 - It is made up of **meristematic tissue (promeristem)** which gives rise to epicotyls and hypocotyl.
 - **Epicotyl** gives rise to **plumule**, which develops into **shoot system** and **hypocotyl** gives rise to **radicle** which develops into **root system**.
 - ii. Embryonal axis:** It is a specialized structure of embryonal axis where two cotyledons are attached.
 - iii. Cotyledon:** There are two cotyledons and they provide nutrition for developing embryo.

Structure and development of Monocot embryo

- The development of monocot embryo is almost similar to that of dicot embryo and is referred to as **Sagittaria type**.
- Here zygote divides transversely producing a two celled proembryo with a **basal cell** and a **terminal cell**.
- The basal cell enlarges to form **vesicular suspensor** cell and the terminal cell gives rise to the whole of the embryo.
- The monocot embryo consists of embryonal axis which is attached laterally to a single shield shaped cotyledon called **scutellum**.
- The portion of the embryonal axis above the level of attachment of scutellum is called **epicotyl** and the portion below the level of cotyledon is called **hypocotyl**.
- Epicotyl** has the **plumule (shoot apex)** covered by a sheath called coleoptile.
- The **hypocotyl** has the **radicle** and **root cap** enclosed in an undifferentiated sheath called **coleorrhizae**.



Differences between Dicot and Monocot Embryo

Dicot Embryo	Monocot Embryo
Basal cell forms 6 – 10 celled suspensors	Basal cell produces single celled suspensor
Terminal cell produces embryo except the radicle	Terminal cell forms the whole of the embryo
The first division of terminal cell is longitudinal	The first division of terminal cell is transverse.
It has two cotyledons	It has single cotyledon
Plumule is terminal and lies in between the two elongated cotyledons	Plumule appears lateral due to excessive growth of single cotyledon.

Differences between Epicotyl and Hypocotyl

Epicotyl	Hypocotyl
It is the part of embryonal axis in between plumule and cotyledonary node.	It is the part of embryonal axis in between cotyledonary node and radicle.
The terminal end of epicotyl is plumule.	The terminal end of hypocotyl is radicle.
In hypogeal germination, epicotyl elongates so that cotyledons remain in the soil.	In epigeal germination, hypocotyl elongates so that cotyledons come out of the soil.

Differences between Coleoptile and Coleorrhiza

Coleoptile	Coleorrhiza
It is a foliar structure enclosing epicotyl bearing shoot apex and leaf primordia.	It is a sheath enclosing radicle and root cap.
It has a terminal pore for the emergence of first leaf	It is a solid structure. No presence of pore.
It protects the plumule during emergence from the soil.	It does not protect the radicle during its passage into the soil.
After emergence during germination, it becomes green and does photosynthesis.	It does not come out of the soil. It remains non green.

Differences between Integument and Testa

Integument	Testa
It is the covering of ovule.	It is the outer covering of seed.
It is thin, one or two layered.	It is thick and single layered.
Cells are living.	Cells are dead.
Sclereids are absent.	Cells are rich in sclereids.
It arises from chalazal end of ovule.	It is derived from outer integument of ovule after fertilization.
It is a pre fertilized structure	It is post fertilized structure

Differences between Perisperm and Pericarp

Perisperm	Pericarp
It is a part of the seed	It is a part of fruit
It is unused nucellus in the seed	It is an outer covering of fruit that developed from ovary wall
It is usually dry	It is dry or fleshy
It is non functional	It is protective covering and also helps in dispersal
It is present in only few seeds	It is found in all fruits

Structure of Monocot seed

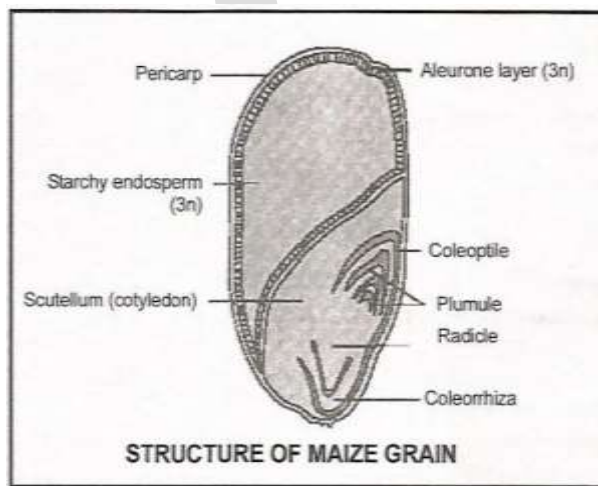
All monocotyledonous seeds consist of the following parts

a. Embryonal axis:

- It is made up of meristematic tissue gives rise to plumule and radicle.
- Around **plumule** fine hair like structures are derived and are called **coleoptile** which gives rise to shoot system (pseudo stem).
- Around the **radicle** fine hair like structures are formed and are called coleorrhizae which gives rise to root system (fibrous root system).

b. Embryonal node:

- There is only one embryonal node in which a single cotyledon is attached to embryonal axis.
- In monocot seed cotyledon is associated with endosperm through **aleurone layer**.

**Seed**

- A seed is defined as fertilized ovule formed inside the fruit.
- A typical seed consists of seed coat and an embryo.

- The **two integuments** of the ovule become the **seed coat**. The **outer integument** forms **testa** while the **inner integument** develops into **tegmen**.
- The micropyle remains as a small pore in the seed coat and it facilitates the entry of oxygen and water into the seed during germination.
- Within the seed coat there is an embryo. It consists of embryonal axis to which one or two cotyledons are attached. The embryo may enter into a state of inactivity called **Dormancy**.
- The embryo is surrounded by a nutritive tissue called **endosperm**.
- Depending upon the presence and absence of endosperm the seeds are classified into two types. They are
 1. Seeds which retain some amount of endosperm even at maturity as it is not completely used up by the developing embryo are called **endospermic seeds** or **albuminous seeds**.
E.g.: - Maize, Sunflower, Castor, barley, wheat
 2. Seeds which do not retain and have no residual endosperm at maturity as it is completely consumed by the developing embryo are called **non-endospermic seeds** or **ex-albuminous seeds**.
E.g.: - Pea, Bean, Groundnut.
- Occasionally in some seeds like black pepper, beet and coffee, remnants of nucellus are also persistent. This residual persistent nucellus or the degenerated part of nucellus of ovule is called **perisperm**.

NOTE

- In a few species the seeds lose viability (able to germinate) within a few months.
- Seeds of a large number of species live for several years. Some seeds can remain alive for hundred years.
- There are several records of very old yet viable seeds.
- The oldest is that of a **Lupine, *Lupinus arcticus*** excavated from **Arctic Tundra**. The seed germinated and flowered after an estimated record of **10,000 years** of dormancy.
- A recent record of **2,000 years** old viable seed is of the **Date palm, *Phoenix dactylifera*** discovered during the archaeological excavation at **King Herod's palace** near **Dead Sea**.

Importance of Seeds

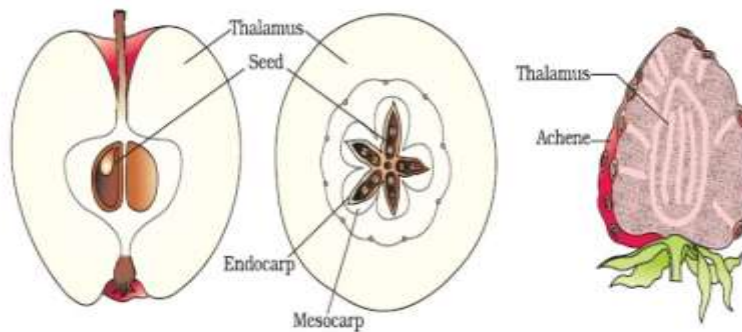
1. **Dependable method:** Pollination and fertilization of seed plants are independent of water. These two reproductive processes are free from requirement of water, hence seed formation is more dependable.
2. **Perennation:** Seed is dry with dormant embryo and thick protective seed coat. It is most suitable for perennation through unfavourable periods.
3. **Dispersal:** Seeds have better adaptive strategies to get dispersed to new habitat and help the species to colonise in other areas.
4. **Reserve food:** Seeds have sufficient quantity of reserve food for nourishing the young seedlings.
5. **Protection:** Hard seed coat provides protection to young embryo.
6. **Variations:** As they are formed due to sexual reproduction, they produce plants with new variations and better genetic recombination. Variations are essential for adaptability to diverse environmental conditions.
7. **Agriculture:** Seed is the basis of agriculture. They can be used as food throughout the year and also to raise crop in the next season.

Fruit

- A fruit is a matured or ripened ovary or Fruit is a post fertilization product of ovary with or without seeds.
- It is formed by the act of pollination and fertilization
- **Parts of fruit:** a fruit consists of two parts a) Fruit wall b) seeds

- **Fruit wall:** The wall of the ovary develops into the wall of fruit called **Pericarp**.
- **Seed:** One or more seeds are present inside the fruit wall.
- **True fruit:** The fruit may be fleshy as in Guava, orange, Mango or may be dry in groundnut, mustard.
- The fruit that develops from the ovary is called **True fruit** or **Eucarp**. **E. g:** - Mango, tomato, lemon, orange.
- **False fruit:** The fruit that develops not only from the ovary but also from other parts of the flower like tepals, bracts, thalamus is called **False fruit** or **Pseudo fruit** or **Pseudocarp**. **E.g:** - Apple, strawberry, cashew nut

False fruits of Apple & Strawberry



Parthenocarpic fruit:

- In certain plants fruit may be formed without the act of pollination and fertilization. Such fruits are called **parthenocarpic fruits** or **seedless fruits** and the phenomenon is called **parthenocarpy**. **E.g.:** - Banana
- The term parthenocarpy was given by **Noll (1902)**. Development of fruits without fertilization is called parthenocarpy and such fruits are called parthenocarpic fruits. Such fruits are seedless. Important parthenocarpic fruits are banana, papaya, guava, pineapple, apple, grapes etc,
- The wall of the ovary develops into fruit wall called **pericarp**. It consists of three layers namely, **outer epicarp, middle mesocarp** and **inner endocarp**.
- The fruit may be fleshy as in Guava, orange, Mango or may be dry in groundnut, mustard.
- The fruit that develops from the ovary is called **True fruit** or **Eucarp**. **E. g:** - Mango, tomato, lemon, orange.
- The fruit that develops not only from the ovary but also from other parts of the flower like tepals, bracts, thalamus is called **False fruit** or **Pseudo fruit** or **Pseudocarp**. **E. g:** - Apple, strawberry, cashew nut
- **Fruits are classified into 3 main types based on the condition of gynoecium. They are**
 - a. A fruit which develops from simple or compound ovary is called **simple fruit**.
E. g: - Mango, Orange
 - b. A fruit develops from polycarpellary, apocarpous gynoecium and is called **Aggregate fruit**.
E. g: - custard apple
 - c. A fruit develops from the entire inflorescence is called **Composite fruit** or **Multiple fruit**.
E. g: - Jack fruit, Pineapple.

Additional Points for Competitive Examination

Parthenocarpy is of two types

1. Stimulative – Fruits are formed as a result of stimulation by pollination but without fertilization. **E.g.:** - certain orchids, grapes etc

2. Vegetative – Seedless fruits can develop without the stimulus of pollination.

E.g.: - Citrus, banana, pineapple, pear, fig etc

Parthenocarpy can also be induced artificially by hormonal treatment. The two commonly used hormones are auxins and gibberellins. They enable flower to develop seedless fruits

without the stimulation of pollination.

E.g.: - 1. Tomato produce seedless fruits if treated with auxin.

2. Grape – wine and pome produce seedless fruits if treated with gibberellin

- Low temperature, frost and fog can induce parthenocarpy in pea, olive, chillies and tomato.

Apomixis

- The phenomenon of formation of seeds without fertilization is called **Apomixis**.
E. g: - Asteraceae and grasses.
- It is a type of asexual reproduction where new individuals are formed without the formation and fusion of gametes.
- Thus, apomixis is a form of asexual reproduction that **mimics sexual reproduction**.
- It is found in some angiosperms particularly in **Grasses**.
- The term apomixis was coined by **Winkler** in **1908**.
- The seeds formed by apomixis are called **Apomitic seeds**.
- There are several ways of development of apomitic seeds.
E. g: - In citrus and mango embryos are formed directly from a diploid cell outside the embryo sac either the nucellus or the integument.

Significance/advantages of Apomixis

- Production of hybrid seeds is costly and too expensive to the farmers.
- The hybrids plants produce from such seeds do not retain the character but the character segregate in them.
- Hybrid varieties of several crops are being extensively cultivated because of their increased productivity.

Polyembryony

- The phenomenon of having more than one embryo in a seed or occurrence of more than one embryo in a seed is called **Polyembryony**.
E. g: - Orange, Mango, Onion
- More often, as in many Citrus and Mango varieties some of the nucellar cells surrounding the embryo sac dividing, protrude into the embryo sac and develop into the embryos. In such species each ovule contains many embryos.

Question Bank

Two-mark question

1. What is tapetum? Mention its significance.
2. What is xenogamy? Mention its significance.
3. What are cleistogamous flowers? Give an example.
4. What are chasmogamous flowers? Give an example.
5. What is self-incompatibility? What is its significance?
6. What is emasculation? Why it is practiced in artificial hybridization?
7. Distinguish between true fruits and false fruits with examples.
8. What is filiform apparatus? What is their function?
9. What are albuminous and ex-albuminous seeds. Give examples.
10. What is apomixis? Give examples.
11. Mention the stages of embryogenesis in angiosperms.

Three-mark questions

1. With a neat labeled diagram explain the structure of pollen grain.
2. Write four characters of wind pollinated flowers and give two examples.
3. What are out breeding devices? Explain briefly the various out breeding devices in plants.
4. Mention the four characters of insect pollinated flowers and give two examples.
5. Differentiate between microsporogenesis and megasporogenesis.
6. What is artificial hybridization? Explain emasculation and bagging techniques use in artificial hybridization of crop improvement programme.
7. a. What is endosperm?
b. Differentiate between nuclear and cellular endosperm with suitable examples.
8. Briefly explain the structure of a typical; dicotyledonous embryo.
9. Describe how does pollination occurs in Vallisneria.
10. Sketch and label the structure of monocot seed.
11. Draw a neat labelled diagram of dicot embryo.
12. Draw a neat labelled diagram of monocot embryo.

Five marks questions

1. Define pollination. Describe the different types of pollination in angiosperms and mention their significance.
2. With a neat labeled diagram describe the structure of T.S of mature anther (microsporangium).
3. With a neat labeled diagram explain the structure of anatropous ovule.
4. Describe the structure of female gametophyte (embryo sac) in angiosperms with a neat labeled diagram.
5. Describe the phenomenon of double fertilization in angiosperms.
6. With a neat labeled diagram describe the structure of dicot embryo.
7. Describe the structure of monocot embryo with a neat labeled diagram.
8. Mention the characteristics of ornithophilous and entomophilous flowers.
9. Explain the structure of Megasporangium.